



TÜRKİYE AND NETHERLANDS GROUP PROJECT

"Food Waste"



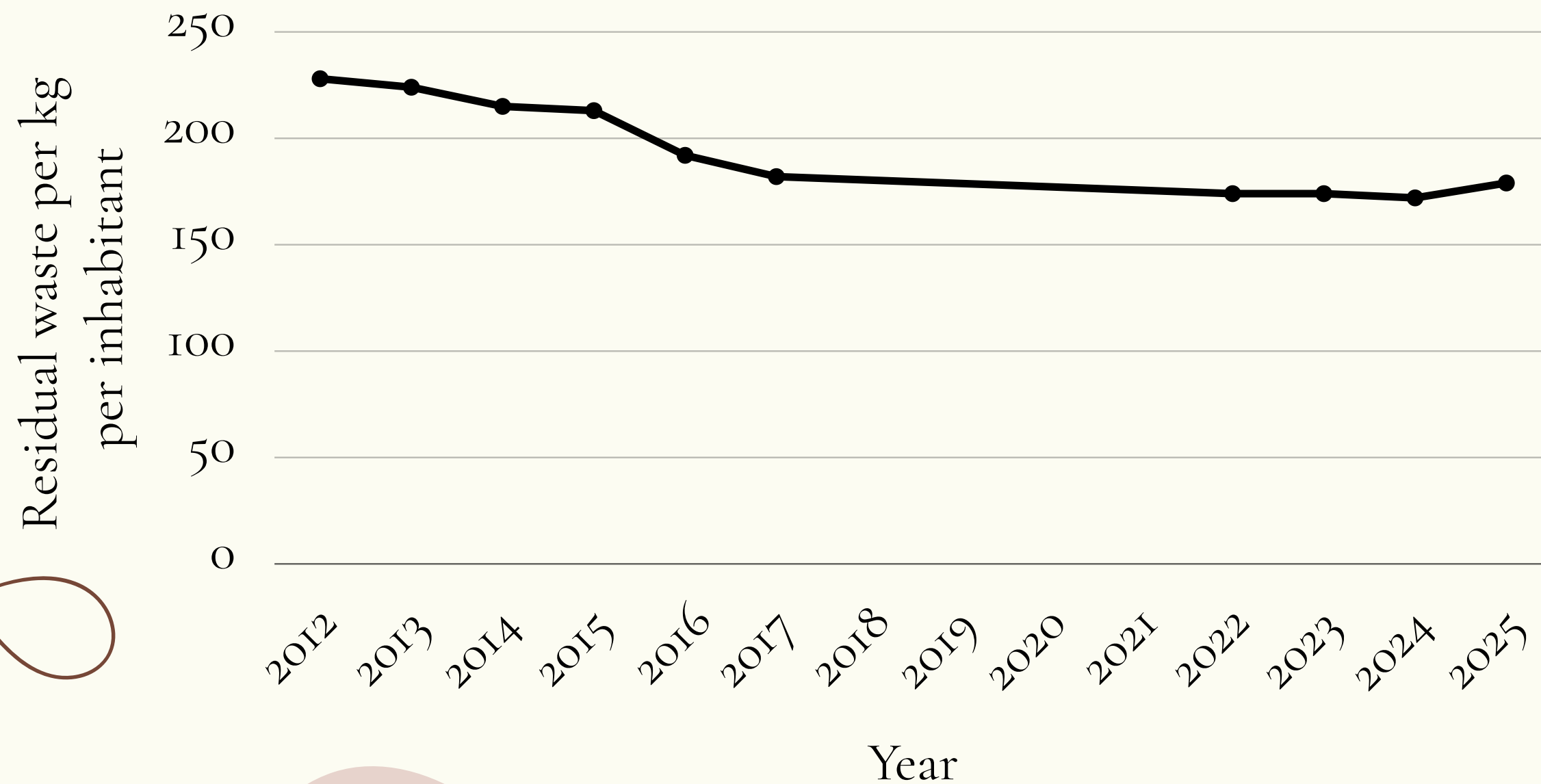
OUR TEAM

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PROBLEM

Food waste is a big problem in the Netherlands. In the Netherlands, approximately 127-161 kg of food per person is lost every year. Food waste is not only a waste but also bad for our climate. Before food reaches the supermarket, it's packaged and transported to your supermarket. This requires gasoline and use of fossil fuels. If you throw the food away, you waste the effort and materials used. Furthermore, food that is thrown away often rots. Rotting food releases methane gas, which contributes to the increased greenhouse effect. People often don't realize that throwing away food has a huge impact on the climate and they also don't realize that it's simply a waste of products

GRAPHIC ALPHEN

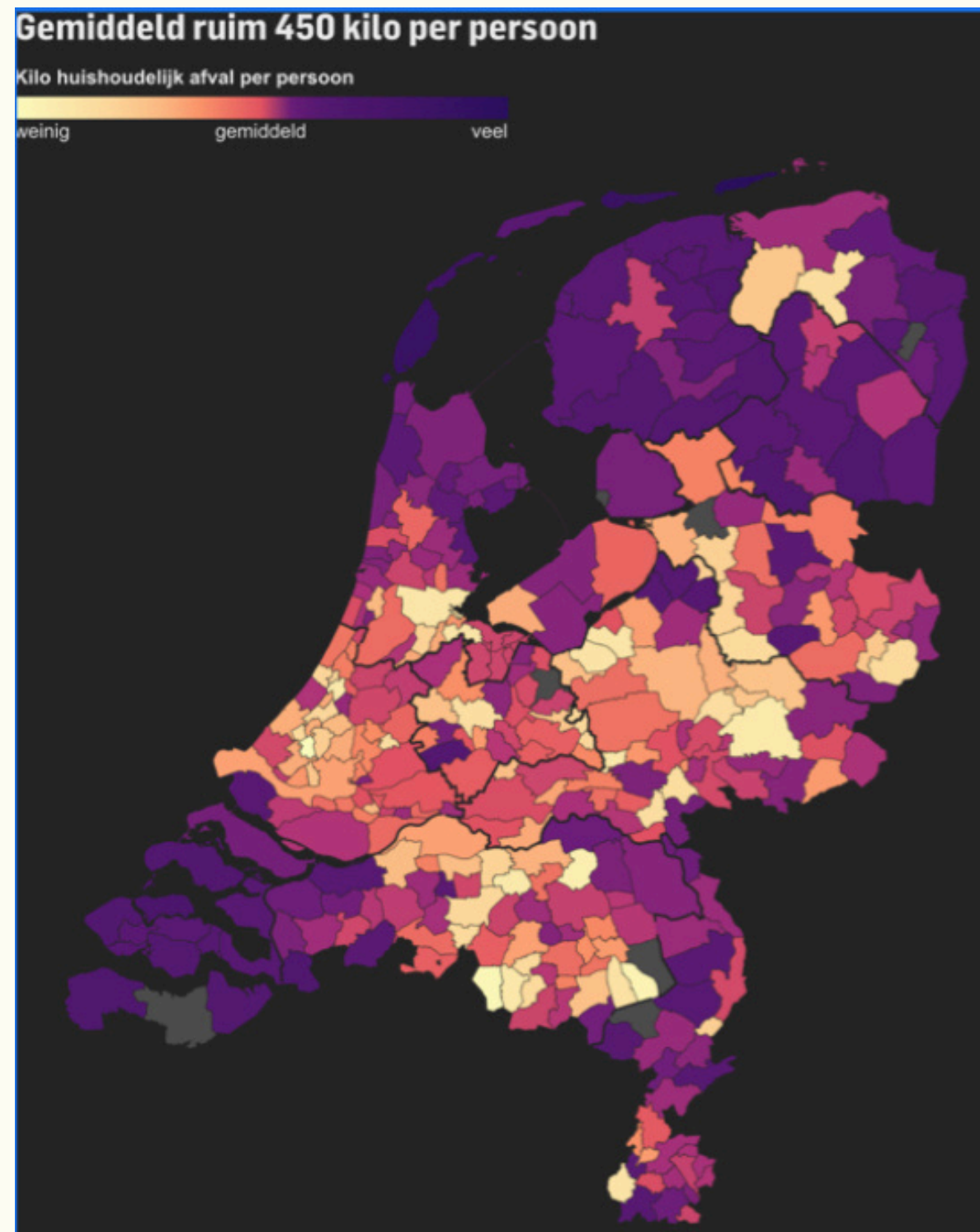


As you can see, we have been wasting less and less, but over the past year there has been a small increase in the amount of food we waste.

this can be caused by the following reasons:

- The number of inhabitants grows
- people are not aware of the consequences of food waste
- people often buy more food than they need
- people get more picky nowadays, the food doesn't only have to taste good but also look good

GRAPHIC ALPHEN



This image shows the entire Netherlands. It shows how many kilos of household waste are thrown away per person per year. You can see that a lot of waste is thrown away in the north, while in the center of the Netherlands, it's less and average. The average is 450 kilos of household waste. This is a huge amount. Some of this waste also consists of food that was actually still good to eat. This means that we are all wasting a huge amount of food, even though it's completely unnecessary.

WHAT IS ALREADY BEING DONE TO COMBAT FOOD WASTE?

IN ALPHEN AAN DEN RIJN

Many different measures are already being taken to combat food waste in Alphen aan den Rijn. For example, there's the food bank, where all expired food is sent so it can be used in another way. Supermarkets also discount products that are about to expire so they can still be sold. Supermarkets also buy limited stock to avoid overstocking and having to throw away half of it because it hasn't been sold. There's also Eetbaar Alphen.

SOLUTIONS

ALPHEN AAN DEN RIJN

- Growing local food (Eetbaar Alphen) so the food doesn't have to travel long and can be harvested when it's needed

Eetbaar Alphen

- Using thrown away food to create new energy, biofuels.
- Discount products in the supermarket that are just past their expiration date or donate them to the food bank

TÜRKİYE

ISTANBUL

Istanbul: A Megacity at the Crossroads of Two Continents

Istanbul is not only Turkey's cultural and economic heart but also one of the largest metropolises in Europe. With an official population of approximately 16 million, which swells to nearly 20 million when accounting for tourism and the daily workforce, the city's scale is truly immense. To put this into perspective for our partners, Istanbul's population is roughly equivalent to that of the entire Netherlands. While this density makes the city a powerful economic engine—contributing over 30% of Turkey's GDP—it also creates a complex logistical challenge. Every day, thousands of tons of food enter the city, and a significant portion of that supply chain results in waste, making Istanbul a critical focal point for sustainability efforts.

TÜRKİYE İSTANBUL

The Invisible Cost: Food Waste in the Metropolis

The food waste generated by Istanbul's massive population represents both a severe environmental crisis and a staggering economic loss. Current data suggests that approximately 1.6 million tons of food are thrown away in Istanbul every year, with the average resident wasting about 102 kilograms of food annually. This figure is not just about the food itself; it represents the wasted water, energy, and labor used from farm to fork. According to data from the Istanbul Commodity Exchange, the annual economic loss reaches approximately \$3.12 billion. For a city striving for modern sustainability goals, addressing this leakage is no longer optional but a financial and ethical necessity.

TÜRKİYE İSTANBUL

Analyzing the Loss: From Households to Retail

When we examine the sources of this waste, we see that the largest share—roughly 60%—comes from households. Istanbul's fast-paced lifestyle, coupled with traditional shopping habits and challenges in portion management, contributes heavily to this domestic waste. However, the problem extends deep into the service industry; restaurants and hotels (particularly the "all-inclusive" concepts) are responsible for about 28% of the total waste. The remaining 12% occurs in the retail sector, where products approaching their expiration dates or those deemed "imperfect" are discarded before even reaching the consumer. This breakdown indicates that solutions must target both individual behavior and industrial systemic changes.



SUGGESTION

Building a Sustainable Bridge: Netherlands-Istanbul Collaboration

The world-leading expertise of the Netherlands in agricultural technology and circular economy principles offers a unique opportunity to solve Istanbul's waste problem. For instance, the hundreds of traditional "neighborhood markets" (semt pazarları) across Istanbul generate tons of organic waste daily that could be converted into compost or biogas through Dutch "upcycling" technologies. Furthermore, combining local digital startups like "Fazla" with established Dutch food banking models could create a streamlined system to rescue surplus food at its source. By merging Istanbul's massive scale with Dutch innovation, we have the potential to create a global success story in urban food management.

COMPARISON

Two Different Worlds, One Common Goal: Istanbul and Alphen aan den Rijn

When we bring these two cities together, we see two different worlds that perfectly complement each other. In Alphen aan den Rijn, the system functions like a small-scale, high-precision watch, where academic rigor, individual awareness, and waste separation discipline are at their peak. In contrast, Istanbul presents a massive surge of energy that demands large-scale management. For instance, while technology in Alphen aan den Rijn is utilized to recycle household waste with maximum efficiency, in Istanbul, that same technology could be used to transform the massive biomass from thousands of daily neighborhood markets into local power plants. While Alphen aan den Rijn teaches us "how to plan with precision," Istanbul proves "how to turn those plans into massive impact" under the most challenging real-world conditions. By merging Istanbul's human capital and logistical power with the technological discipline of the Netherlands, the result will be more than just an environmental project—it will be a revolution in efficiency.

CONCLUSION

"Based on the findings and local observations gathered, it is evident that the recent slight increase in waste in Alphen aan den Rijn—driven by population growth, lack of awareness, and rising aesthetic demands—proves that food waste is a universal challenge of modern life. In a megacity like Istanbul, where the population equals that of the entire Netherlands, these issues manifest on a much more massive scale. Istanbul's annual potential of 1.6 million tons of waste offers the world's largest testing ground for the systemic solutions and technological discipline found in Alphen aan den Rijn. Current statistical trends remind us that reducing waste is not a one-time fix but requires continuous innovation and social awareness. By merging Istanbul's unique dynamism with Dutch methodology, we can build a partnership that protects not only these two cities but the future of food itself. A solution that succeeds in a system as vast as Istanbul will serve as a viable model for the entire world."



THANK YOU !